

Dermoscopy and In Vivo Imaging

Dermoscopy is more sensitive than visual inspection in detecting orifices, vessels, and specific vascular patterns such as crown, radial, flower, and punctiform arrangements.

In vivo confocal microscopy reveals a round, well-circumscribed lesion with central cystic areas filled with brightly refractile material, corresponding to molluscum bodies seen on histopathology.

Laboratory

Laboratory testing is not routinely required for the diagnosis of molluscum contagiosum, as clinical and dermoscopic evaluation is typically sufficient.

In uncertain cases, confirmation can be achieved through histological examination, polymerase chain reaction, or electron microscopy. Histology shows the characteristic "nuts in a sack" pattern. Electron microscopy identifies the brick-shaped virions of molluscum contagiosum virus.

Histopathology

Haematoxylin and eosin staining of molluscum contagiosum lesions typically shows a crateriform, hyperplastic epidermis or endophytic infundibular hyperplasia, forming a circumscribed cup-shaped pseudotumor. This structure contains large cells with granular, eosinophilic intracytoplasmic molluscum bodies (also known as inclusion bodies or Henderson-Paterson bodies). These inclusion bodies can measure up to 35 microns in diameter and consist of

millions of virions that compress the keratinocyte nuclei. They progressively enlarge and become basophilic before being released onto the skin surface.

Unusual histological variants include pseudocystic, giant, and pedunculated forms. Disseminated, confluent, or atypical giant lesions may indicate immunosuppression, particularly HIV infection.

Special stains such as Lendrum's phloxine tartrazine or toluidine blue/Giemsa are rarely needed, as inclusion bodies are usually evident on routine staining. However, these stains may be helpful when folliculitis, abscess formation following follicular rupture, or a dense inflammatory infiltrate obscure the inclusions.

Occasionally, a dense dermal lymphocytic infiltrate with CD30-positive cells may mimic anaplastic large cell lymphoma or lymphomatoid papulosis. Molluscum bodies can be identified by immunohistochemistry on formalin-fixed, paraffin-embedded tissue, and cross-reactivity with Melan A—a melanocytic marker—has been reported.

Immunohistochemical studies are reserved for research and not used in routine diagnosis. Additional histological findings may include perilesional fibro-oedematous to fibromyxoid stroma, and rarely amyloid-like deposits, anetoderma, or metaplastic ossification. Follicular induction in the adjacent epidermis should not be mistaken for basal cell carcinoma.

In approximately 4% of cases, molluscum contagiosum may coexist with other lesions such as epidermal cysts or melanocytic naevi, potentially masking the characteristic changes. Coinfection with other pathogens—such as human papillomavirus or *Cryptococcus neoformans*—within the same lesion can occur, especially in immunocompromised individuals.

Differential Diagnosis

Molluscum contagiosum must be differentiated from other conditions, particularly in the setting of possible sexual transmission.

In immunocompetent patients, mimickers include: - Genital warts (condyloma acuminata) - Flat warts - Lichen planus - Lichen nitidus - Condylomata lata of secondary syphilis - Pyogenic granuloma - Ectopic sebaceous glands - Dermal cyst - Vulvar lymphangioma circumscriptum - Keratoacanthoma - Basal cell carcinoma - Amelanotic melanoma

In immunosuppressed patients, opportunistic infections that may resemble molluscum contagiosum include: - *Penicillium marneffe*i (penicilliosis) - *Cryptococcus neoformans* (cryptococcosis) - *Coccidioides* spp. (coccidioidomycosis) - *Paracoccidioides brasiliensis* - *Aspergillus fumigatus* (aspergillosis) - *Sporothrix schenckii*